The BRITISH UNIVERSITY IN EGYPT Informatics and Computer Science	**25CSCI34H** **Group Project** **2025/2026**
Module Title: **Computer Systems Security**	
Module Leader **Dr. Ghada Elsayed**	Semester **Two**
Assessment Weight **%40 of the total course mark**	Due Date **Due date (exact day to be specified in CW map at the beginning of the semester)**

Instructions to students:

1. This is a group assignment. The group consists of 3 students, and their names are subject to the approval of the module leader.

2. Submission: The submission is via the e-learning system only

3. Assessment: The group project is divided into six graded submission phases, each contributing to the final grade. Peer review is integrated to adjust individual contributions in certain phases. Review the **Deliverables and Weight table** for more details.

4. Feedback: Feedback will be given through discussions

5. You can only submit your own work. Any student suspected of plagiarism will be subject to the procedures set out in the UAR.

6. **Use of AI and Academic Integrity: Students may use generative AI tools to support their learning, such as brainstorming topics, refining research questions, organizing outlines, or improving grammar and clarity. However, AI tools are not allowed to be used to generate substantive content, analysis, or conclusions submitted as the student's own work. Presenting AI-generated text, ideas, data, or solutions without proper acknowledgment constitutes plagiarism, just as it would with any other**

source. All submitted work must reflect the student's independent thinking and understanding, and students must be able to explain and defend their work. Misuse of AI or plagiarism will be treated as a violation of the institution's academic integrity policy.

Assessment overview

The assessment for this module is a group-based project designed to provide students with practical experience in designing, implementing, and evaluating secure computer systems. The project requires students to apply theoretical concepts covered in the modules, such as cryptography, network security, web security, and safe transactions, to a realistic system-level security problem.

Assessment objectives

The objectives of this assessment are to enable students to:

1. **Select a computer system or application to secure**, considering the group members' **majors** (Computer Networks, Information Systems, Artificial Intelligence, or Computer Science) to leverage their expertise. **Interdisciplinary ideas and solutions that combine skills across majors are highly encouraged**.
2. **Choose individual roles** within the group:
    - Backend/Database Security
    - Frontend/API Security
    - DevOps/Infrastructure Security
3. **Identify and review five recent journal or conference papers** relevant to each student's assigned role to inform system design and security decisions.
4. **Develop a project proposal** based on the literature review, including:
    - Proposed security solution(s) for the system
    - Risk and threat analysis
    - Tools, technologies, and interfaces to be used
    - Timeline and task schedule for implementation
5. **Design the system architecture and role-specific modules**, ensuring:
    - Each role has a clearly defined module and interface
    - Security mechanisms are specified per module
    - Potential threats and risks are analysed for each module
6. **Implement and test role-specific modules**, verifying that each component:

    - Meets the security and functionality requirements defined in the proposal
    - Correctly apply cryptography, authentication, access control, or other relevant security mechanisms
    - Mitigates identified threats and risks
7. **Integrate all role-specific modules** into a complete system and perform **integration testing**, ensuring:
    - End-to-end security and functionality
    - Compatibility between modules
    - System-level threat mitigation and robustness
8. **Prepare and deliver a system demonstration and presentation**, highlighting:
    - Individual contributions and role-specific work
    - Security features and their effectiveness
    - Results of functional and integration testing
    - How interdisciplinary approaches contributed to the project

Project Goal

The goal of this project is to develop **individual and group work** that makes use of one or more of the following computer system security topics, while encouraging **interdisciplinary thinking** based on students' specializations:

- **Cryptography**: Symmetric-key encryption, Public-key encryption, Digital Signatures, Secure hash functions, Key certificates, Key management
- **Secure communication protocols**: Secure Sockets Layer / Transport Layer Security (SSL/TLS)
- **Malware**: Identification, analysis, and mitigation
- **Firewalls**: Network and host-based firewall configuration
- **Web Security**: Secure web application design, authentication, authorization, and protection against common web attacks
- **Secure Electronic Transaction (SET)**: Secure handling of electronic payment and transaction processes

Deliverables & Submission Phases (12-Week Schedule)

The group project is divided into **six graded submission phases**, each contributing to the final grade. Peer review is integrated to adjust **individual contributions** in certain phases.

Phase 1: Project Selection & Role Assignment — *Week 2, 7.5%*

Contents:

- Selection of a computer system or application to secure, considering group members' majors (Computer Networks, Information Systems, AI, Computer Science).
- Assignment of individual roles: Backend/Database Security, Frontend/API Security, DevOps/Infrastructure Security.
- Brief rationale for system selection and **interdisciplinary potential**.

Grading Focus:

- Appropriateness of system selection
- Role assignment clarity and alignment with expertise
- Interdisciplinary innovation

Phase 2: Literature Review & Project Proposal — *Week 4, 17.5%*

Contents:

- **Individual:** Review of **5 recent journal/conference papers** relevant to each student's role.
- **Group:** Project proposal including:
    - Proposed security solution(s)
    - Risk/threat analysis
    - Tools, technologies, and interfaces
    - Timeline and task schedule

Grading Focus:

- Depth, relevance, and critical analysis of literature review
- Feasibility, clarity, and completeness of the project proposal
- Quality of risk/threat analysis
- Well-defined tools/interfaces and integration plan

Phase 3: Role-Specific Design & Implementation — *Week 7, 15%*

Contents:

- Design of **role-specific modules** with clear interfaces.
- Implementation of modules applying security principles (cryptography, access control, web security, etc.).
- **Functional testing per role** to validate module security and functionality.
- Role-specific documentation detailing design, implementation, and testing.

Grading Focus:

- Secure and robust module design
- Correct application of security mechanisms
- Functional testing coverage and results
- Quality and clarity of role-specific documentation

Phase 4: System Integration & Testing — *Week 9, 15%*

Contents:

- Integration of all role-specific modules into a complete system.
- End-to-end **integration testing**, verifying security and functionality.
- Updated system architecture documentation reflecting integration.
- Identification and resolution of system-level issues.

Grading Focus:

- Successful integration of all modules
- Effective mitigation of system-level threats
- Robustness, compatibility, and end-to-end functionality
- Integration testing documentation clarity and completeness

Phase 5: Final Report — *Week 11, 20%*

<u>Final Report:</u>

1. Length: 1300 - 1800 words
2. Format: Typed using word processing software.
3. Content: Complete report (with citations and references) documenting all the work completed. All parameters, graphs, and values must be clearly labeled and identified.
4. Discussion: The submission of the final report will be composed of a discussion that the module leader and the TA will administer.

Contents:

- Comprehensive documentation of the project, including:
    1. Executive summary
    2. System description & architecture
    3. Risk & threat analysis
    4. Role-specific design and implementation
    5. Functional testing results per module
    6. Integration testing results
    7. Interdisciplinary contributions
    8. Lessons learned and reflections
    9. References & appendices (diagrams, code snippets, etc.)

Grading Focus:

- Completeness and clarity of the report
- Evidence of individual role contributions
- Quality of risk/threat analysis and mitigation
- Integration of interdisciplinary ideas
- Professional presentation of results

Phase 6: Demo & Presentation — *Week 12, 25%*

Contents:

- **System demonstration** highlighting security features.
- Group presentation covering:
    - Individual contributions
    - Role-specific implementation
    - Risk/threat mitigation
    - Interdisciplinary aspects of the solution
- Q&A assessing understanding of the system and security decisions.

Grading Focus:

- Functional system demonstration
- Clear, professional, and engaging presentation

- Participation of all group members
- Depth of understanding during Q&A

Peer Review

- Conducted **after Phase 5**.
- Each student rates the contribution of other group members on:
    - Design & implementation contributions
    - Timeliness and quality of work
    - Collaboration and communication
    - Initiative and problem-solving
- Peer review results **adjust individual marks** for Phases 3–6 (10% of the individual grade).

Summary Table: Deliverables and Weight

Phase	Week	Weight	Deliverable
1	2	7.5%	Project selection, role assignment, interdisciplinary rationale
2	4	17.5%	Literature review (individual 15%), project proposal (group 2.5%)
3	7	15%	Role-specific design & implementation, functional testing, documentation (individual).
4	9	15%	System integration, integration testing, and updated documentation (group)
5	11	20%	Final report: comprehensive documentation, interdisciplinary insights, lessons learned (group)
6	12	25%	Demo & presentation, Q&A (individual assessment 20%/ group work 5%)
Peer Review	After Phase 5	10%	Adjust individual contributions for Phases 3–6

<u>Important Remarks</u>

- Due Date: Will be announced.
- Upload your implementation and final report to e-learning in a single ZIP file attachment.

- Students HAVE TO attend the presentation & demo. If a student is absent, they will be considered absent for the Discussion, for the final report and software, and the Presentation and Demo deliverables.
- Projects will not be accepted without submitting proposals beforehand for approval.
- **Use of AI and Academic Integrity: Students may use generative AI tools to support their learning, such as brainstorming topics, refining research questions, organizing outlines, or improving grammar and clarity. However, AI tools are not allowed to be used to generate substantive content, analysis, or conclusions submitted as the student's own work. Presenting AI-generated text, ideas, data, or solutions without proper acknowledgment constitutes plagiarism, just as it would with any other source. All submitted work must reflect the student's independent thinking and understanding, and students must be able to explain and defend their work. Misuse of AI or plagiarism will be treated as a violation of the institution's academic integrity policy.**

Module Specification – (Programme Specs Ver. 6.0)

The BRITISH UNIVERSITY IN EGYPT

Module Code: CSCI34H	**Title:** Computer Systems Security	
Level: H	**Modular weight:** 10	**Faculty/Dept:** ICS
Pre-requisite modules: CSCI03I – Operating Systems, CSCN01I - Introduction to Computer Networks & Data Communication		
Reassessment: No restriction		
Module Leader: Dr. Ghada Elsayed		
Semester taught: Two.		
Date of latest revision: March 2022		

Aims

To provide students with a deep understanding of the security issues associated with computer systems. The emphasis is on the underlying theory, mechanisms and techniques for implementing security, to develop a critical comprehension of the fundamentals of secure specification and design.

Intended Learning Outcomes

On completion of this module, students should be able to:

Knowledge and Understanding

1. Classify algorithms and techniques used to secure computer systems. [A11]

Intellectual Skills

2. Analyze how cryptographic algorithms work and how they are applied to provide secure Internet connections. [B5]
3. Critically assess a website or a client system and identify security problems. [B12]

Practical and Professional Skills

4. Apply cryptography algorithms and techniques. [C12]
5. Discover any risks or safety aspects that may be involved in the operation of computer networks. [C9]

General and Transferable Skills

6. Plan self-learning and performance improvement as the foundation for ongoing professional development. [D8]

Employability

This module will provide opportunities for students to:

1. Understand the importance of adaptability to ensure their ongoing effectiveness in an ever-changing environment. [A1]
2. Carry out a range of complex ICT activities related to their work that involve application software. [B.3.1]
3. Keep up to date with developments in own areas of professional specialisation. [C.1.1]
4. Demonstrate effective planning, prioritisation, and organisation to plan activities and carry them through effectively. [C.2.2]

Indicative Content

- Computer security principles
- Cryptography: Symmetric-key encryption, Public-key encryption, Digital Signatures, Secure hash functions, key certificates, key management
- Secure Sockets Layer (SSL/TLS)
- Malware
- Firewalls
- Web Security
- Secure Electronic Transaction (SET)

Methods of Learning, Teaching, and Assessment

Total student effort for the module: 100 hours on average over one semester.

Type of session	ILOs Covered	Typical Student Effort		
		Typical number of semester/s	Typical hours per week	Total hours
Lecture	1-3	12	2	24
Tutorial	-			
Laboratory	4-6	12	2	24
Private Study	1-6			52

Assessment

Assessment Type	Weight %	ILOs Assessed	Exam Semester	Exam/ Written Coursework Length
Group project (up to 3 students) with a report. The students are then discussed individually. The group work and individual parts deserve 50% each.	40	4-6	1	1300 - 1800 words
Unseen written exam	60	1-3	1	2-hour exam

Methods of Feedback

In response to assessed work:

- Feedback will be provided for each assessed component in oral form as appropriate. If students require additional feedback, they are welcome to speak with the TA and the module leader. Brief written feedback on the examination will appear on the module eLearning page after results are published.
- Generic exam feedback will be given on the e-learning system.

Developmental feedback generated through teaching activities:

- Dialogue between module-leader and students during lectures, consultation by the module-leader during project-phases, and discussion during Labs and workshops between staff and students.

Indicative Reading List

- Goodrich and Tamassia, Introduction to Computer Security, Addison-Wesley, 2010.
- Kaufman, Perelman, and Speciner, Network Security: Private Communication in a Public World, Second Edition, Prentice Hall, 2002.
- Stallings, W., “Network Security Essentials”, 4th Edition, Pearson (2011).
- Stallings, W., “Cryptography and Network Security Principles and Practice”, 5th Edition, Pearson (2011).